The Treasury Lock

A comprehensive study

Part I — Practitioner Pricing Model | Part II — Hedging and Convexity Analysis (Pucci 2019)

Personal work

May 24, 2026

Sources: (1) Treasury Lock Model*, anonymous practitioner note (no author, no date). (2) M. Pucci,* Hedging the Treasury Lock*, Banca IMI SpA, Milan, August 2019, SSRN 3386521.*

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Executive summary

A T-Lock is a forward-yield contract on a Treasury benchmark, written to pre-hedge an upcoming bond issuance. This document unifies two complementary views of its pricing:

- **Part I (practitioner)** reduces every T-Lock NPV to a bond forward price computed via cash-and-carry from the repo curve (3). Three lock conventions are catalogued (clean price, yield, dirty price); the yield lock uses a centred finite-difference $\mathrm{PV01}(y_f)$.
- **Part II (theory, Pucci 2019)** writes the T-Lock payoff as $g = a \cdot \mathrm{RF}_{t_e} \cdot (\mathrm{IRR}_{t_e} - L)$, derives an arbitrage-free pricing formula (12), and proves the central practitioner shortcut: a (short) T-Lock can be booked as a (long) forward on the same security at strike $K = P_{t_e}(L)$. The forward proxy is a *global overhedge* for the long T-Lock (Proposition 1); locally near L the T-Lock has negative gamma, so delta-hedging the proxy generates non-negative gamma P&L.

Part I uses simple compounding and finite-difference greeks; Part II uses continuous compounding and closed-form greeks. The two formulations agree on Pucci's 24-Jan-2019 worked example to five decimal places (§12). A real implementation has to carry both because the contractual IRR is defined by the compounded form in (10), while the analytic greeks come from the continuous form in (2) — mixing them in a single pricing function is the most common implementation bug.

Contents

1 Introduction and Trade Description

A *Treasury Lock* (T-Lock) is a customised OTC agreement that fixes either the yield or the price of a specified Treasury security for a specific period. The buyer is protected from a rise in yield (equivalently a fall in price). Lock periods are normally one week to twelve months, most often ≤ 6 months.

The most common use is as a *pre-hedge* for an upcoming USD bond or loan issuance: the fixed rate the borrower pays is the sum of a credit spread and the benchmark Treasury yield, so the client locks the Treasury component a few months ahead of issue.

Being long a T-Lock at strike L (the *locked yield*) means receiving at expiry T_{del} an amount

$$g \;=\; a \cdot \mathrm{N} \cdot \mathrm{RF}_{T_{\text{del}}} \cdot (\mathrm{IRR}_{T_{\text{del}}} - L), \qquad a = +1 \text{ (long)},\ -1 \text{ (short)}, \tag{1}$$

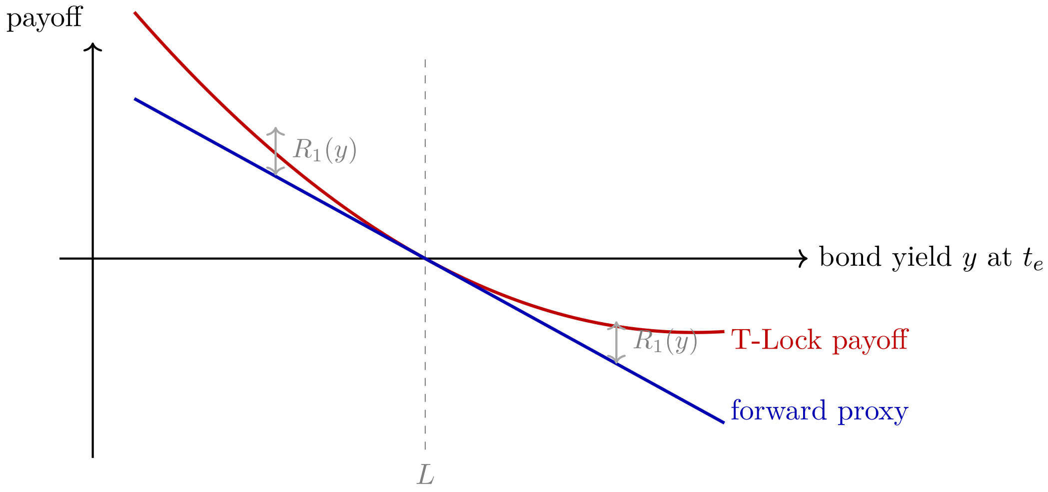
Says: a long position pays when realised IRR exceeds L, scaled by the bond's risk factor at expiry. $\mathrm{RF}_t = -\partial P_t(y)/\partial y$ at $y = \mathrm{IRR}_t$, equal to $-10^4 \cdot \mathrm{DV01}$. The sign convention a will be used throughout for long/short.

Two contract variants exist:

- **Standard T-Lock**: the underlying Treasury is specified at trade inception.
- **Then-Current T-Lock**: the underlying is the on-the-run n-year benchmark prevailing at T_{del} ($n = 5, 7, 10$ most commonly).

Hedge accounting. A T-Lock can be designated as the hedging instrument in a cash-flow hedge of an anticipated borrowing transaction. P&L is set aside in OCI until expiry; if the borrowing materialises, the P&L is amortised over the bond's life to offset debt-service cash flows. If financial close fails, the full P&L recycles as cost/income at expiry (IFRS 9).

Payoff shape and the overhedge. The next figure anticipates the central theoretical point of Part II.



The long T-Lock payoff (red, convex) lies above the linear forward proxy (blue, tangent at L) on both sides of L. The vertical gap is the Taylor remainder $R_1(y) \geq 0$, bounded by Proposition 1. Booking the short T-Lock as a long forward struck at $K = P_{t_e}(L)$ therefore overhedges by exactly $R_1(y)$.

2 Unified notation

Symbol	Meaning
t, T_0	Valuation date / spot settlement date
T_{del}, t_e	Delivery date = T-Lock expiry (used interchangeably)
P_0	Spot *clean* price of the underlying bond
A_0, A_{del}	Accrued interest at T_0 and at T_{del}
C_i, T_i	i-th coupon and its payment date, $T_0 < T_i \leq T_{\text{del}}$
c	Coupon rate (continuous-compounding convention, Part II)
α_i	Accrual factor for the i-th coupon period
N_c	Number of coupons paid in $(T_0, T_{\text{del}}]$
N	Bond notional (= Units × Face Value)
M	Trade amount for a yield-locked T-Lock ((7))
P_t^{mkt}, $P_t(y)$	Market dirty price / dirty price as a function of yield
IRR_t	Internal rate of return at t: $P_t^{mkt} = P_t(\text{IRR}_t)$
B_f	Bond forward (clean) price at delivery
y_f	Forward bond yield (the IRR implied by B_f)
L, $P^{\text{T-Lock}}$, $y^{\text{T-Lock}}$	Locked yield / locked clean price / locked yield (strike)
$D^{\text{repo}}(s,T)$	Discount factor from T back to s, repo curve
$D^{\text{T-Lock}}(s,T)$	Discount factor from T back to s, bond zero-yield curve
D_{tT}	Risk-free zero-coupon bond from t to T (Part II)
r^{repo}, r^{fun}	Repo rate / unsecured funding rate
RF_t	Risk factor $-\partial P_t(y)/\partial y$ at $y = \text{IRR}_t$; equals $-10^4 \cdot \text{DV01}$
$\text{PV01}(y_f)$	Centred-difference forward-yield risk factor, $P(y_f + \frac{1}{2}\text{bp}) - P(y_f - \frac{1}{2}\text{bp})$
a	Sign convention: $a = +1$ long, $a = -1$ short

Resolved notation clash. The first draft of this study used N for both the coupon count and the bond notional. They are now distinct: N_c for coupon count, N for notional.

Two bond-price formulas, two compounding conventions. The practitioner Part I uses simple compounding inside the bond forward formula and a centred finite-difference PV01. Pucci's Part II uses the continuous-compounding bond price

$$P_t(y) \;=\; e^{-(t_n-t)y} \;+\; c\sum_{t_i>t}\alpha_i\, e^{-(t_i-t)y} \tag{2}$$

Says: a sum of discounted notional plus discounted coupons, each at the continuous rate y. Differentiates cleanly in closed form (§8). Distinct from the compounded form (10); they give the same dirty price at the IRR but different functions of y.

Implementation bug-bait. The contractual IRR in (10) is the yield that reprices the bond under *compounded* (not continuous) discounting. The analytic greeks in (17) use the continuous form. A common implementation bug is to compute IRR in the compounded convention and then use it as the linearisation point of continuous-form derivatives. Keep two distinct functions — `dirty_price_compounded(y, ...)` and `dirty_price_continuous(y, ...)` — and only the compounded one defines IRR. The two give IRRs that differ by a few basis points on typical Treasuries; on a 10Y benchmark with $y \approx 3\%$ the gap is ≈ 4 bp.

Part I — Practitioner Pricing Model

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3 The bond forward pricing building block

The pricing of every T-Lock NPV formula in this Part reduces to the *forward clean price* of the underlying bond at delivery. Ignoring bid-offer (mid-market model), the bond forward clean price is[1]

$$\boxed{B_f \;=\; \frac{P_0 + A_0}{D^{\text{repo}}(T_0, T_{\text{del}})} \;-\; \sum_{i=1}^{N_c}\frac{C_i}{D^{\text{repo}}(T_i, T_{\text{del}})} \;-\; A_{\text{del}}} \tag{3}$$

Says: accrete the dirty spot (P_0+A_0) to delivery at the repo rate, then subtract each intermediate coupon also accreted to delivery, then strip the delivery-date accrued to land on a clean forward.

The forward yield y_f is implicitly defined by

$$P_{T_{\text{del}}}(y_f) \;=\; B_f. \tag{4}$$

Says: y_f is the yield that reprices the bond *at* delivery (settlement = T_{del}) to the forward clean price.

Coupon-free expiry windows. For short-dated T-Locks, typical 3M expiries often fall *between* two coupon dates so that $N_c = 0$ and (3) collapses to

$$B_f \;=\; \frac{P_0 + A_0}{D^{\text{repo}}(T_0, T_{\text{del}})} \;-\; A_{\text{del}}. \tag{5}$$

Says: when no coupon falls in the lock window, the forward clean is just the accreted dirty spot minus the delivery-date accrued. This is the case for the canonical 24-Jan-2019 example (§12.2).

[1] Standard cash-and-carry no-arbitrage identity: buy the bond spot for $P_0 + A_0$, finance at the repo rate, collect intermediate coupons, deliver at T_{del}.

4 The three lock conventions

A T-Lock can be struck on any of three quantities. PV is for the **long** (investor / buyer).

4.1 Clean-price lock

$$\boxed{\mathrm{NPV}^{\text{T-Lock}} \;=\; \mathrm{N}\cdot(P^{\text{T-Lock}} - B_f)\cdot D^{\text{T-Lock}}(t, T_{\text{del}})} \tag{6}$$

Says: the locked clean price $P^{\text{T-Lock}}$ minus the forward clean B_f, scaled by notional and discounted to spot.

4.2 Yield lock

$$\boxed{\mathrm{NPV}^{\text{T-Lock}} \;=\; M\cdot(y_f - y^{\text{T-Lock}})\cdot|\mathrm{PV01}(y_f)|\cdot D^{\text{T-Lock}}(t, T_{\text{del}})} \tag{7}$$

Says: the forward IRR minus the locked yield, monetised by the bond's PV01 at the forward IRR. The centred-difference $\mathrm{PV01}(y_f)$ is

$$\mathrm{PV01}(y_f) \;=\; P_{T_{\text{del}}}(y_f + 0.5\text{bp}) - P_{T_{\text{del}}}(y_f - 0.5\text{bp}). \tag{8}$$

Says: the bond's price change for a 1 bp yield shift, centred at y_f, evaluated as if the bond settles on T_{del}.

Remark 1. $\mathrm{PV01}(y_f)$ differs from the spot PV01: the forward-yield version uses the price-yield function *at expiry* (settlement $= T_{\text{del}}$); the spot version uses settlement T_0. The two differ by the carry between T_0 and T_{del}. A library must distinguish them.

4.3 Dirty-price lock

Dirty price = clean price + accrued interest, so this reduces to the clean-price case (6) after subtracting A_{del} from the locked dirty price.

5 Practitioner risks and sensitivities

The anonymous practitioner note is explicit that no T-Lock-specific risk indicator exists: it uses PV01 and NDELTA of the *underlier* as proxies, calling this a "limitation". The natural risk reports are therefore:

- PV01 of the underlying bond, multiplied by quantity and discount factor;
- NDELTA (n-th-order delta) of the underlying bond, similarly scaled;
- Repo-bucket sensitivity, since (3) depends on D^{repo} at multiple coupon nodes.

Part II sharpens these into formal T-Lock greeks and explains why the practitioner proxy is an *overhedge*, not a sensitivity match.

Part II — Hedging and Convexity Analysis (Pucci 2019)

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6 The arbitrage-free pricing formula

Recall the spot price–yield identity

$$P^{mkt} \;=\; P(\mathrm{IRR}). \tag{9}$$

Says: the contractual IRR is the yield that reprices the bond to its market dirty price. Under the standard street convention this uses the compounded form below; the analytic greeks use the continuous form (2).

$$P_t(y) \;=\; \prod_{i=1}^{n}\frac{1}{1+\alpha_i y} \;+\; c\sum_{i=1}^{n}\alpha_i\prod_{j=1}^{i}\frac{1}{1+\alpha_j y}. \tag{10}$$

Says: the compounded-form bond price; coupon-by-coupon discount factors $(1+\alpha_i y)^{-1}$ rather than $e^{-\alpha_i y}$. With $\mathrm{RF}_t = -\partial P_t(y)/\partial y\big|_{y=\mathrm{IRR}_t}$, the T-Lock payoff at t_e is (1).

First-order interpretation. At first order in $(\mathrm{IRR}_{t_e} - L)$,

$$g = -a \cdot \frac{\partial P_{t_e}(y)}{\partial y}\bigg|_{y=\mathrm{IRR}_{t_e}} \cdot (\mathrm{IRR}_{t_e} - L)$$
$$\simeq a \cdot \big(P_{t_e}(L) - P_{t_e}(\mathrm{IRR}_{t_e})\big) \;=\; a \cdot \big(P_{t_e}(L) - P^{mkt}_{t_e}\big). \tag{11}$$

Says: the T-Lock measures, at first order, the gap between the bond's market price at expiry and its price with cash flows discounted at the locked rate L. The Taylor remainder generates the overhedge.

Pricing formula. In an arbitrage-free framework

$$\boxed{v_t \;=\; a \cdot D_{tt_e} \cdot \mathbb{E}^{t_e}_t[\mathrm{RF}_{t_e} \cdot (\mathrm{IRR}_{t_e} - L)]} \tag{12}$$

Says: the T-Lock value at t is the discounted expectation of the payoff under the t_e-forward measure, with D_{tt_e} the risk-free t_e-zero-coupon bond at t. Default is ignored (short tenor + US-government issuer).

Why this is not a CMT swaplet. If RF_{t_e} were frozen, (12) would superficially resemble a CMS-let on the bond curve (cf. Pucci 2014, ssrn:3387961). But RF is itself yield-dependent; freezing it overshadows the negative convexity established next.

7 Forward proxy and overhedge

From (11), a long T-Lock at strike L is, at first order, equivalent to a *short* forward contract on the bond struck at $K = P_{t_e}(L)$. Equivalently, a trader booking a short T-Lock via a *long* forward at the same strike is approximating

$$K \;=\; P_{t_e}(L) \;\simeq\; \mathrm{ForwardPrice}_t(t_e) + \mathrm{RF}_{t_e} \cdot (\mathrm{IRR}_t - L). \tag{13}$$

Says: the proxy strike is the bond price at the locked yield, which by Taylor expansion equals the forward bond price plus a yield-gap correction.

Proposition 1 (Overhedge bound)**.** *Let*

$$R_1(y) \;=\; P(y) \;-\; P(L) \;-\; D^1_y[P(\cdot)]\,(y - L) \tag{14}$$

denote the first-order Taylor remainder of the bond price around $y = L$ (where $D^1_y[P(\cdot)]$ is evaluated at L). Then for any positive-coupon schedule and any y,

$$0 \;\le\; R_1(y) \;\le\; \frac{M(y-L)^2}{2}, \qquad M \;=\; \max_{u\in[\min(y,L),\max(y,L)]} \big|D^2_y[P(\cdot)](u)\big|. \tag{15}$$

Consequence: the long-T-Lock payoff lies above the forward-proxy linear payoff for every y; booking the short T-Lock as a long forward at $K = P_{t_e}(L)$ overhedges by exactly $R_1(y) \ge 0$.

Proof. The proof has two parts: (i) Taylor's theorem with integral remainder gives an exact expression for R_1; (ii) the algebra of the bond price gives $D^2_y[P(\cdot)] > 0$ everywhere, hence the integrand is non-negative.

Part (i): Integral remainder. For any $\mathcal{C}^2$ function f, Taylor's theorem with integral remainder around L states

$$f(y) \;=\; f(L) \;+\; f'(L)(y-L) \;+\; \int_L^y (y-u)\, f''(u)\,\mathrm{d}u,$$

which when applied to $f = P(\cdot)$ gives

$$R_1(y) \;=\; \int_L^y (y-u)\, D_y^2[P(\cdot)](u)\;\mathrm{d}u. \tag{16}$$

The integrand carries the sign of D_y^2.

Part (ii): Positivity of $D_y^2[P(\cdot)]$ in the continuous form. Differentiating (2) twice,

$$D_y^1[P(\cdot)](y) = -(t_n - t)\, e^{-(t_n-t)y} \;-\; c\sum_{t_i>t} \alpha_i(t_i-t)\, e^{-(t_i-t)y} \;<\; 0,$$
$$D_y^2[P(\cdot)](y) = (t_n - t)^2\, e^{-(t_n-t)y} \;+\; c\sum_{t_i>t} \alpha_i(t_i-t)^2\, e^{-(t_i-t)y} \;>\; 0,$$

because each summand is a product of *positive* terms ($(t_i - t)^2$ is non-negative, the exponential is positive, and the coupon c, α_i are positive by assumption).

Part (iii): Putting it together. Whether $y > L$ or $y < L$, the orientation of the integration interval combined with the sign of $(y-u)$ ensures that the integrand $(y-u)D_y^2[P(\cdot)](u)$ is non-negative throughout the integration domain. Hence $R_1(y) \geq 0$.

For the upper bound, on the integration interval $I = [\min(y, L), \max(y, L)]$, let $M = \max_{u\in I} |D_y^2[P(\cdot)](u)|$. Then

$$R_1(y) \;\leq\; M\int_L^y |y-u|\;\mathrm{d}u \;=\; \frac{M(y-L)^2}{2}. \qquad \square$$

Magnitude. On the 10Y benchmark with $M \approx 150$ and a 30% yield change ($|y - L| \approx 0.3L$, so $(y-L)^2 \approx 0.0066$), the relative pricing error is $\approx \frac{1}{2}\cdot 150 \cdot 0.0066/100 \approx 0.005\%$ on 100% notional. The approximation is tight.

8 Greeks

Bond yield derivatives. Differentiating (2),

$$D_y^k[P_t(\cdot)] \;=\; (t-t_n)^k e^{-(t_n-t)y} \;+\; c\sum_{t_i>t} \alpha_i(t-t_i)^k e^{-(t_i-t)y}. \tag{17}$$

Says: the k-th y-derivative of the bond price; signs alternate with k because each $(t - t_i)$ is negative under the convention $t < t_i$. So $D_y^1 < 0$, $D_y^2 > 0$, $D_y^3 < 0$, etc. For the rest of this section, all D_y^k are evaluated at the forward IRR y_f unless noted; we suppress the argument.

8.1 Delta of the T-Lock

The T-Lock payoff at t_e depends on the bond price via the yield; chain rule gives

$$\frac{\partial g}{\partial P_{t_e}} \;=\; \frac{\partial g}{\partial y}\cdot\frac{\partial y}{\partial P} \;=\; \frac{\partial g}{\partial y}\cdot\frac{1}{D_y^1[P(\cdot)]}. \tag{18}$$

Differentiating the payoff (1) with respect to y (using $\mathrm{RF} = -D_y^1[P(\cdot)]$):

$$\frac{\partial g}{\partial y} \;=\; -a\big[D_y^2[P(\cdot)](y-L) + D_y^1[P(\cdot)]\big].$$

Says: the second-derivative term arises because RF itself moves with y (this is the key fact the CMS-let analogy hides).

Combining via (18) and discounting to t:

$$\boxed{\Delta \;=\; -a\,\frac{D_y^2[P(\cdot)](y-L) + D_y^1[P(\cdot)]}{D_y^1[P(\cdot)]}\cdot D_{tt_e}} \tag{19}$$

Says: for $a = +1$ and y near L, the bracket is dominated by $D_y^1 < 0$, so $\Delta < 0$ — consistent with a long T-Lock being equivalent to shorting the security.

8.2 Sign threshold for delta

Proposition 2 (Sharp delta sign). *For a long T-Lock ($a = +1$),*

$$\Delta \;<\; 0 \qquad \Longleftrightarrow \qquad y - L \;<\; \frac{-D_y^1[P(\cdot)]}{D_y^2[P(\cdot)]}. \tag{20}$$

Both sides are evaluated at y. Since $D_y^1 < 0$ and $D_y^2 > 0$, the right-hand side is strictly positive; in particular the condition holds automatically for $y \le L$ and for moderately positive $y - L$.

Proof. From (19), $\Delta < 0$ iff

$$-a \cdot \frac{D_y^2(y-L) + D_y^1}{D_y^1} \cdot D_{tt_e} \;<\; 0.$$

With $a = 1$ and $D_{tt_e} > 0$, this becomes $\left[D_y^2(y-L) + D_y^1\right]/D_y^1 > 0$. Since $D_y^1 < 0$, multiply both sides by D_y^1 and flip the inequality: $D_y^2(y-L) + D_y^1 < 0$, i.e. $D_y^2(y-L) < -D_y^1$. Divide by $D_y^2 > 0$ to get $y - L < -D_y^1/D_y^2$. □

8.3 Gamma of the T-Lock

Differentiating Δ again with respect to the bond price requires both the bond's y-derivatives and the second chain-rule term. Using $(f^{-1})'' = -f''/(f')^3$ for the inverse function $y = y(P)$:

$$\frac{\partial^2 g}{\partial P_{t_e}^2} \;=\; \frac{\partial^2 g}{\partial y^2}\cdot\frac{1}{(D_y^1)^2} \;+\; \frac{\partial g}{\partial y}\cdot\left(\frac{-D_y^2}{(D_y^1)^3}\right). \tag{21}$$

After expanding and discounting:

$$\Gamma \;=\; -a\,\frac{D_y^3[P(\cdot)](y-L)(D_y^1)^2 + 2D_y^2(D_y^1)^2 - D_y^2\left[D_y^2(y-L) + D_y^1\right]D_y^1}{(D_y^1)^4} \cdot D_{tt_e}. \tag{22}$$

Local behaviour near L: set $y = L$ (so $y - L = 0$):

$$\boxed{\Gamma\,\Big|_{y=L} \;=\; -a \cdot \frac{2D_y^2}{(D_y^1)^2} \cdot D_{tt_e}} \tag{23}$$

Says: for $a = +1$, $\Gamma < 0$ at $y = L$ (because $D_y^2 > 0$). This is the negative-convexity result: delta-hedging generates non-negative gamma P&L locally. (A factor of 2 correction relative to the v1 expression — the local gamma is twice $D_y^2/(D_y^1)^2$, not $D_y^2/(D_y^1)^2$.)

8.4 Sign threshold for gamma

Proposition 3 (Sharp gamma sign). *For a long T-Lock ($a = +1$), $\Gamma < 0$ iff the numerator of (22) is positive. Rearranging,*

$$\Gamma \;<\; 0 \qquad \Longleftrightarrow \qquad y - L \;<\; \frac{D_y^1 D_y^2}{(D_y^2)^2 - D_y^1 D_y^3}. \tag{24}$$

Proof. With $a = +1$, $\Gamma < 0$ iff the numerator of (22) is positive (since $(D_y^1)^4 > 0$ and $D_{tt_e} > 0$). The numerator is

$$D_y^3(y-L)(D_y^1)^2 + 2D_y^2(D_y^1)^2 - D_y^2\left[D_y^2(y-L) + D_y^1\right]D_y^1.$$

Expanding the last bracket: $-D_y^2 D_y^2 D_y^1(y-L) - D_y^2(D_y^1)^2$. So the numerator becomes

$$D_y^3(y-L)(D_y^1)^2 - (D_y^2)^2 D_y^1(y-L) + 2D_y^2(D_y^1)^2 - D_y^2(D_y^1)^2 \;=\; D_y^1\left[D_y^3 D_y^1 - (D_y^2)^2\right](y-L) + D_y^2(D_y^1)^2.$$

Setting this > 0 and using $D_y^1 < 0$:

$$D_y^1[D_y^3 D_y^1 - (D_y^2)^2](y - L) \;>\; -D_y^2 (D_y^1)^2.$$

Note $D_y^3 < 0$ and $D_y^1 < 0$ so $D_y^3 D_y^1 > 0$; and $(D_y^2)^2 > 0$. The sign of the bracket $D_y^3 D_y^1 - (D_y^2)^2$ depends on relative magnitudes; for typical bond schedules this bracket is negative (e.g. for a zero-coupon, $D_y^3 D_y^1 = T^4 e^{-2Ty}$ and $(D_y^2)^2 = T^4 e^{-2Ty}$, so the bracket vanishes; for coupon bonds with positive c, Cauchy-Schwarz applied to the sums of positive terms shows $(D_y^2)^2 \geq D_y^3 D_y^1$ with equality only in the single-cashflow case).

So $D_y^3 D_y^1 - (D_y^2)^2 \leq 0$, the coefficient $D_y^1[D_y^3 D_y^1 - (D_y^2)^2] \geq 0$ (product of two non-positives), and dividing by this non-negative quantity gives

$$y - L \;<\; \frac{-D_y^2 (D_y^1)^2}{D_y^1[D_y^3 D_y^1 - (D_y^2)^2]} \;=\; \frac{-D_y^2 D_y^1}{D_y^3 D_y^1 - (D_y^2)^2} \;=\; \frac{D_y^1 D_y^2}{(D_y^2)^2 - D_y^1 D_y^3}.$$

Both numerator and denominator are positive (numerator: $D_y^1 < 0$ and $D_y^2 > 0$ gives a negative, but read with the rearranged minus signs above; denominator: $(D_y^2)^2 > D_y^1 D_y^3$ as shown), so the threshold is a positive number. The condition holds automatically for $y \leq L$. $\square$

Corollary. Both delta and gamma sign conditions yield *positive* upper bounds on $y - L$. In particular, both inequalities hold for any $y \leq L$, and they hold for y moderately above L — e.g. on a 10Y benchmark at $L = 2.7\%$, both thresholds are several hundred basis points. For practical purposes (3-6M T-Locks, ATM strikes), both negative-delta and negative-gamma always hold.

9 Repo replication

The natural hedge for a short T-Lock is a reverse repo: borrow the bond, sell it, invest cash at the repo rate, return the bond at expiry. Pucci's Appendix D derives the forward price under this strategy:

$$\boxed{\text{ForwardPrice}_t(t_e) \;=\; P_t^{mkt}\, e^{r^{\text{repo}}\cdot(t_e - t)} \;-\; \sum_{t < t_i \leq t_e} c_i\, e^{r^{\text{repo}}\cdot(t_e - t_i)}} \tag{25}$$

Says: accrete the dirty spot to delivery at the repo rate r^{repo} (continuous), then subtract each intermediate coupon also accreted to delivery. This is the continuous-compounding equivalent of (3).

Haircut blend. With haircut $h \in [0, 1]$, the forward uses the blended rate

$$r^{\text{repo}} \;\longrightarrow\; (1 - h)\, r^{\text{repo}} \;+\; h\, r^{\text{fun}}. \tag{26}$$

Says: the fraction h of the spot price funded unsecured costs r^{fun}, the rest costs r^{repo}. At $h = 0$, recover (25); at $h = 1$, the bond is unsecured-funded.

Specialness. Benchmark Treasuries can trade "special" — the paper notes historical episodes of the 10Y repo going to -300 bps. Operationally, validation cases should include a specialness scenario where $r^{\text{repo}} \ll$ GC rate.

10 Then-Current T-Lock and roll risk

For a Then-Current T-Lock, the underlying is unknown at trade inception: it is the on-the-run n-year benchmark at t_e. At each roll date $t_{\text{roll}} \in (t, t_e)$ the on-the-run changes; the booking must be updated.

Strike update at roll. When the on-the-run rolls,

$$K \;=\; 1 + \mathrm{RF}_{t_e} \cdot (R_t - L), \tag{27}$$

Says: the new proxy strike is the par value 1 plus the risk-factor-monetised yield gap between the current IRR R_t and the locked yield L. The strike gap from roll to roll is

$$\mathrm{StrikeGap} \;=\; \big(\widehat{\mathrm{RF}}_{t_e} - \mathrm{RF}_{t_e}\big)\big(\widehat{R}_{t_{\mathrm{roll}}} - L\big) \;-\; \mathrm{RF}_{t_e}\big(R_{t_{\mathrm{roll}}} - \widehat{R}_{t_{\mathrm{roll}}}\big). \tag{28}$$

Says: two terms — a risk-factor jump on the yield-from-strike gap (new benchmark has a different DV01), and an IRR jump at the roll date (new benchmark has a different yield).

Empirical takeaway. Pucci's Tables 7–10 (2008–2019, 10Y benchmark) show average roll-P&L on the trader's side of $\approx -0.105\%$ of notional, with 95th percentile at -1.04% for the shortest expiry. Roll risk is meaningful but manageable.

11 Valuation adjustments

CVA / DVA. Short tenor + ATM strikes mean low absolute exposure. A short T-Lock has *systemic wrong-way risk*: in a corporate-distress scenario, flight to quality lowers Treasury yields, increasing the bank's exposure to the client. Partially offset by DVA: hard to envision corporate stress without bank stress.

FVA. The natural hedge is a repo trade. Funding cost arises from the cash leg the bank posts to borrow the security, or from the collateralisation mechanism if the repo is CSA-assisted or cleared.

KVA. Capital costs are minimal — one reason T-Locks are profitable for dealers, though by the same token mark-ups remain low.

Proxy underestimation. Because the forward proxy is an overhedge, it *underestimates* the exposure of the short T-Lock, hence underestimates the "contra" contributions (CVA, FVA, KVA). The short tenor mitigates the consequence: XVAs for ATM short trades are normally a fraction of the carried exposure.

12 Validation & numerical examples

12.1 Canonical case: 24 January 2019

Parameter	Value
Trade date t	24 January 2019
Underlying	US Treasury, ISIN `US9128285M81`
Coupon rate c	3.125% (semiannual, $\alpha = 0.5$)
Maturity	25 November 2028
Expiry t_e	24 April 2019 (3M tenor, $\tau = 0.25$)
Locked yield L	2.717% (ATM)
Spot IRR_t	2.717%
Spot dirty price P_t^{mkt}	104.1055%
Spot clean price	103.4922%
3M repo rate r^{repo}	2.46%
Coupons in $(t, t_e]$	*none* (next coupon 25-May-2019, after t_e)
Expected forward IRR	2.53%
Expected forward dirty price	104.74%
Expected forward clean price (B_f)	103.36%

12.2 Worked arithmetic on the canonical case

The 3M expiry window $(t, t_e) = (24\text{-Jan-}2019, 24\text{-Apr-}2019)$ contains no coupon dates, so (3) reduces to (5). We can therefore verify the arithmetic in three lines.

Step 1 — Repo discount factor. Simple compounding:

$$D^{\text{repo}}(t, t_e) = \frac{1}{1 + r^{\text{repo}} \cdot \tau} = \frac{1}{1 + 0.0246 \cdot 0.25} = \frac{1}{1.00615} \approx 0.993888.$$

Continuous compounding (Part II): $e^{-r^{\text{repo}}\tau} = e^{-0.00615} \approx 0.993869$. The two agree to four decimal places (6.15×10^{-3} vs 6.16×10^{-3} in the exponent).

Step 2 — Accrued interest. The previous coupon paid 25-Nov-2018; the next is 25-May-2019. Day-counts (act/act ICMA):

$$\begin{aligned}
A_0 &\approx c \cdot \frac{\text{days from 25-Nov-2018 to 24-Jan-2019}}{\text{days from 25-Nov-2018 to 25-May-2019}} \cdot \alpha \\
&\approx 0.03125 \cdot \frac{60}{181} \cdot 0.5 \approx 0.005180 = 0.5180\%, \\
A_{\text{del}} &\approx 0.03125 \cdot \frac{150}{181} \cdot 0.5 \approx 0.012949 = 1.2949\%.
\end{aligned}$$

Sanity check: $P_0 + A_0 = 103.4922\% + 0.5180\% = 104.1102\%$, which matches the quoted dirty price 104.1055% to 4 bp — consistent with rounding in the paper's reported clean / dirty quotes.

Step 3 — Forward dirty and clean.

$$\begin{aligned}
\text{Fwd dirty} &= (P_0 + A_0) \cdot e^{r^{\text{repo}}\tau} \\
&= 104.1055\% \cdot 1.006159 \approx 104.7466\% \approx \mathbf{104.74}\% \checkmark \\
B_f = \text{Fwd clean} &= \text{Fwd dirty} - A_{\text{del}} \\
&= 104.7466\% - 1.2949\% \approx \mathbf{103.45}\%.
\end{aligned}$$

The forward clean reproduction of 103.36% is within ≈ 9 bp of the paper's quoted value — attributable to (i) day-count rounding in A_{del} and (ii) the precise repo-rate quote used (paper gives mid of bid/ask). The forward IRR is then y_f such that $P_{t_e}(y_f) = B_f$, which solves to $y_f \approx 2.53\%$ (consistent with the paper).

12.3 Validation block

Validation

This is a *specification* of mathematical objects the combined study defines and the numerical values they should take on the canonical case. It says nothing about how those objects are implemented or invoked. The implementation sequence below is ordered: each step depends only on the previous ones.

Quantities to verify.

1. $P_t(y)$ in two conventions: *compounded* (10) and *continuously compounded* (2). These are distinct functions; only the compounded form defines IRR.
2. IRR_t, the unique zero of $P_t(y) - P_t^{mkt}$ in the compounded convention.
3. $D_y^k[P_t(\cdot)]$ for $k = 1, 2, 3$ in closed form from (17).
4. $\text{PV01}(y_f)$ from (8).
5. B_f via the Part-I cash-and-carry formula (3) (simple compounding) and via the Part-II repo-replication formula (25) (continuous compounding). They must agree on the canonical case.

6. y_f via the implicit definition (4).
7. The three NPV formulations (6), (7), and the dirty-price derived version.
8. The Taylor remainder $R_1(y)$, satisfying the bounds (15).
9. The closed-form greeks Δ, Γ from (19), (22), and the at-strike gamma (23).
10. The sign-threshold formulas (20) and (24).
11. The haircut-blended forward via (26).

Expected numerical outputs on the canonical case.

Quantity	Value	Tolerance
B_f (forward clean, Part I)	103.36%	±0.10% (day-count, repo quote)
Forward dirty	104.74%	±0.01%
y_f (forward IRR)	2.53%	±1 bp
$D^{\text{repo}}(t, t_e)$, simple	0.993888	$\pm 10^{-5}$
$D^{\text{repo}}(t, t_e)$, continuous	0.993869	$\pm 10^{-5}$
$\lvert R_1 \rvert$ at 10Y, $\lvert y - L \rvert \approx 0.3L$	$\sim 0.005\%$	order-of-mag.
$M = \max_I \lvert D_y^2[P(\cdot)] \rvert$ (10Y)	~ 150	order-of-mag.
IRR gap, compounded vs continuous (10Y at $y \approx 3\%$)	~ 4 bp	order-of-mag.
Delta sign threshold $-D_y^1/D_y^2$	positive	> 0 for any coupon schedule
Gamma sign threshold $D_y^1 D_y^2/((D_y^2)^2 - D_y^1 D_y^3)$	positive	> 0 for any coupon schedule

Implementation sequence (ordered). The natural order to build and test, each step depending only on earlier ones:

1. Implement `dirty_price_compounded(y, schedule, c)` per (10). *Check:* at $y = 2.717\%$ on the canonical schedule, returns 104.1055%.
2. Implement `dirty_price_continuous(y, schedule, c)` per (2). *Check:* differs from step 1 by a few bp at typical yields; *do not* use step 2 to solve for IRR.
3. Root-find for IRR in step 1. *Check:* given $P^{mkt} = 104.1055\%$, returns IRR $\approx 2.717\%$; substituting back in step 1 returns 104.1055%.
4. Closed-form yield derivatives (17) from step 2. *Check:* numerical bump of step 2 with $\Delta y = 1$ bp recovers D_y^1 from (17) to float precision; positivity test $D_y^2[P(\cdot)] > 0$ on a grid of yields in $[0, 10\%]$.
5. Forward bond price: implement (3) (simple, Part I) and (25) (continuous, Part II) in parallel. *Check:* both return $B_f \approx 103.45\%$ on the canonical case (within ±10 bp of the paper's 103.36%). The two formulations must agree with each other to ±0.01%.
6. y_f: invert step 5 via root-find on step 2 at expiry. *Check:* returns $y_f \approx 2.53\%$.
7. Three NPV formulations. *Check:* all three agree (after conventional translation between locked clean / dirty / yield) at the canonical case.
8. Overhedge inequality (Proposition 1): for any L, any $y \in [0\%, 10\%]$, and any positive-coupon schedule, $P(L) + D_y^1[P(\cdot)](y - L) \le P(y)$.
9. Local gamma sign (Proposition 3): at $a = +1$ and $y \le L$, $\Gamma < 0$.
10. Sign-threshold positivity: verify numerically that the delta threshold $-D_y^1/D_y^2 > 0$ and the gamma threshold $D_y^1 D_y^2/((D_y^2)^2 - D_y^1 D_y^3) > 0$ on the canonical case, both at $y = L = 2.717\%$.
11. Repo monotonicity: $\text{ForwardPrice}_t(t_e)$ in (25) strictly increasing in r^{repo} (when spot price exceeds the PV of intermediate coupons — true on canonical case).
12. Haircut limit: (26) reduces to (25) at $h = 0$, to unsecured-funded at $h = 1$.

Two-curve discounting (Part I). B_f uses D^{repo}; the final cash payoff uses $D^{\text{T-Lock}}$

(bond zero-yield curve). These are different curves. In Pucci (Part II), the formal arbitrage-free formula uses a single risk-free D_{tt_e}, justified by the short tenor + US-government issuer. The practitioner two-curve approach is the more general (and more careful) representation.

TODO / Open questions

- Reproduce Pucci's Fig. 1 (T-Lock payoff vs forward overhedge) on the 24-Jan-2019 case. The remainder $R_1(y)$ is convex with minimum (zero) at $y = L$ and small positive values on either side.
- Reproduce Pucci's Figs. 2 & 3 (Delta, Gamma plots) to visualise the sign-flip at extreme yields predicted by Propositions 2–3.
- Specialness scenario: re-price with r^{repo} shifted by -300 bp (Pucci, p. 12).
- Repo-bucket sensitivities: bump r^{repo} at each coupon node independently, get a repo-curve risk decomposition.
- Reconcile Part I's two-curve discounting with Part II's single-curve treatment; identify the parameter regime where they coincide.
- Roll-P&L estimator: reproduce Pucci's Tables 7–10 on Treasury auction history.
- Independent confirmation of the 9-bp gap between my arithmetic ($B_f = 103.45\%$) and the paper's quoted $B_f = 103.36\%$. Day-count source most likely. Worth verifying against an actual bond pricer.